

Compiler-Emitted Nearest-Neighbor Systolic Arrays on FPGA

A Kung-style LU and one-sided Jacobi SVD case study, benchmarked against Vitis HLS

MS Thesis Proposal · Gary Pham · Drexel ECE

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1 Motivation

FPGA systolic arrays remain one of the most efficient substrates for dense linear algebra, but designers still hand-write RTL for each kernel or accept the area/latency overhead of high-level synthesis. Polyhedral-to-systolic compilers such as AutoSA [?] have shown that schedules can be generated automatically, yet the emitted arrays typically use long-range interconnect and non-streaming interfaces, which complicates system integration on modern AMD/Xilinx platforms where AXI4-Stream is the lingua franca [?, ?].

This thesis argues for the opposite constraint set: *strict nearest-neighbor connectivity and AXI4-Stream-compliant outputs by construction*. The claim is that this constraint pair, once baked into a shared PE template, is expressive enough to cover several important dense-linear-algebra kernels while yielding place-and-route results competitive with, or better than, Vitis HLS on the same C source.

2 Prior Work Positioning

FPGA implementations of SVD are dense in the literature. The Brent–Luk–Van Loan mesh [?] underlies most subsequent two-sided designs, including Hemkumar & Cavallaro’s complex-matrix CORDIC array [?], Ma et al.’s FPGA SVD processor [?], and Ahmedsaid & Amira’s improved systolic array [?]. One-sided (Hestenes-Jacobi) variants trade the two-sided rotation for better parallelism on rectangular matrices; representative work includes Wang & Zambreno’s floating-point Hestenes-Jacobi core [?], Karthikeyan’s one-bit-rotation CORDIC PEs [?], Kalaycıoğlu’s real-time two-sided design [?], the UCSB scalable engine with Maximum-Data-Sharing ordering [?], and a recent data-stream Jacobi architecture [?]. LU decomposition on systolic arrays traces back to Kung & Leiserson [?] and has similarly wide coverage.

The novelty of this thesis is **not** another SVD or LU accelerator. It is the compiler-plus-template flow, with LU and SVD serving as demonstrators driven from a shared PE template.

3 Contribution and Thesis Statement

A nearest-neighbor AXI4-Stream PE template, combined with a schedule-generation flow, can emit both a Kung-style LU decomposer and a one-sided Jacobi SVD engine from a single hardware template, matching or exceeding Vitis HLS on area $\times$ latency for both kernels on a Kintex-class target.

Concretely, the thesis will deliver:

1. A parameterized nearest-neighbor PE template with an AXI4-Stream boundary that respects UG1399 handshake semantics.
2. A schedule generator that maps LU and one-sided Jacobi SVD onto $N \times N$ meshes of that template.

3. Post-place-and-route benchmarks against Vitis HLS on identical C reference and against the closest prior FPGA-SVD numbers.
4. An open-source release of the flow, sufficient for reproducibility.

4 Repository Status (as of proposal date)

Progress already in the working tree, ahead of the winter-term start:

- `kung_lu_decom/` — VHDL Kung-style LU on a 4×4 nearest-neighbor mesh with divide/multiply-add/register cells and a fixed-point divider. Simulation clean.
- `kung_lu_support/` — portable C harness that generates stimulus, extracts L/U from the hardware output stream, and verifies $A = L \cdot U$. Runs via `make run-sim`.
- `kung_svd/` — 8×8 one-sided Jacobi SVD prototype: 18-bit Q1.16 PE with Givens rotation (`svd_pe.vhd`), vectoring-mode CORDIC angle engine (`cordic_rot.vhd`), column-pair Jacobi controller with an FSM `IDLE` $\rightarrow$ `PAIR` $\rightarrow$ `CORDIC` $\rightarrow$ `BROADCAST` (`jacobi_ctrl.vhd`), configurable `ROWS` $\times$ `COLS` array (`svd_array_core.vhd`), and an AXI4-Stream wrapper (`svd_axi_stream.vhd`). Structural testbench compiles and runs.

Outstanding work before publishable results: (a) numerical golden-reference check on the SVD output stream (currently the testbench prints but does not assert), (b) Vitis HLS baselines on the target board, and (c) post-place-and-route numbers.

5 Approach

5.1 PE template

The current template exposes N/E/S/W AXI4-Stream ports with `TVALID`/`TREADY`/`TLAST`, a small local register file, and a parameterized arithmetic core. LU uses a divide/subtract/multiply core (`Div_cell.vhd`, `MA_cell.vhd`, `M_cell.vhd`, `R_cell.vhd`). SVD uses a Givens-rotation MAC PE plus a shared CORDIC angle engine broadcast from the array edge.

5.2 SVD case study

The 8×8 grid runs a one-sided Jacobi schedule: the controller selects column pairs $(0, 1), (2, 3), \dots$ then $(1, 2), (3, 4), \dots$, feeds pair data to a 16-iteration vectoring-mode CORDIC to obtain (c, s) , and broadcasts them to all PEs for the sweep. Fixed-point precision is Q1.16 in 18-bit words. The first milestone is a numerically-verified 4×4 SVD; the second is 8×8 (matching the largest size widely reported in prior two-sided work [?, ?]).

5.3 Baseline and comparators

Comparator	Reference	Metric
Vitis HLS (same C, same board)	Internal baseline	LUT/FF/BRAM/DSP, $F_{\max}$, cycles
Ma et al. FPGA SVD processor	[?]	Reported cycles & area, normalized
UCSB scalable Jacobi engine	[?]	Throughput vs. bandwidth
DSB-Jacobi (data-stream)	[?]	BRAM footprint, streaming latency

6 Six-Month Schedule (Jan 27 – Jul 24, 2027)

Weeks	Milestone	Deliverable
1–3	Numerical golden reference for SVD; Vitis HLS baselines for LU and 4×4 SVD	Baseline table (LUT/FF/BRAM/DSP, $F_{\max}$, cycles) checked into docs/
4–7	Fix Jacobi controller edge cases; convergence verification on 8×8 random matrices	Passing testbench with $\ A - U\Sigma V^T\ $ error plots
8–12	First post-P&R numbers on Kintex-7 for both LU and 8×8 SVD	Area/latency table vs. Vitis HLS baseline
13–16	Extract shared PE/schedule template so LU and SVD emit from one flow	Compiler flow write-up; SVD scale-up study
17–20	Full benchmark sweep; paper draft (target: FCCM/FPL 2027 or TRETTS)	Submitted paper draft; thesis chapters 1–4 in review
21–24	Thesis defense preparation; PhD application writing sample finalized	Defended thesis, tagged v1.0 open-source release

7 Risks and Mitigations

- **CORDIC accuracy at fixed-point.** Mitigation: 16-bit fraction starting point per Ma et al. [?]; widen only if reconstruction error exceeds 10^{-3} on 8×8 test matrices.
- **Convergence of the one-sided sweep.** Mitigation: the current design uses a fixed 8-sweep schedule; add off-diagonal-norm tracking to promote to an adaptive stop if convergence is uneven.
- **Scope creep into rectangular matrices or floating-point.** Mitigation: keep the thesis to square, fixed-point Q1.16; document Hestenes-Jacobi rectangular support as future work.
- **Board unavailability.** Mitigation: primary target Kintex-7 (already in the lab); simulation-only fallback path exists via the current xsim flow.
- **Vitis HLS producing surprisingly good baselines.** This is an acceptable outcome; the paper still stands if the compiler flow matches HLS with a fraction of the DSP count or with strictly nearest-neighbor placement, which HLS does not guarantee.

8 What I Am Asking For

1. Your read on whether this field — compiler-emitted nearest-neighbor systolic arrays with LU/SVD as demonstrators — is still workable as a thesis contribution given the density of prior FPGA-SVD work, or whether you would steer me toward an adjacent framing.
2. Confirmation of scope for a six-month MS thesis under your advisement.
3. Target-device sign-off (proposed: Kintex-7 already available; open to Zynq UltraScale+ if you prefer for the paper).
4. Agreement to co-author the resulting FCCM/FPL/TRETTS submission.
5. A short kickoff meeting before winter term so weeks 1–3 land on the ground running.

References

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Prepared by Gary Pham for review by Prof. Prawat Nagvajara. Repository: `/Users/garypham/Drexel/FPGA`. This document is a working proposal; scope and schedule will be revised after our kickoff meeting.